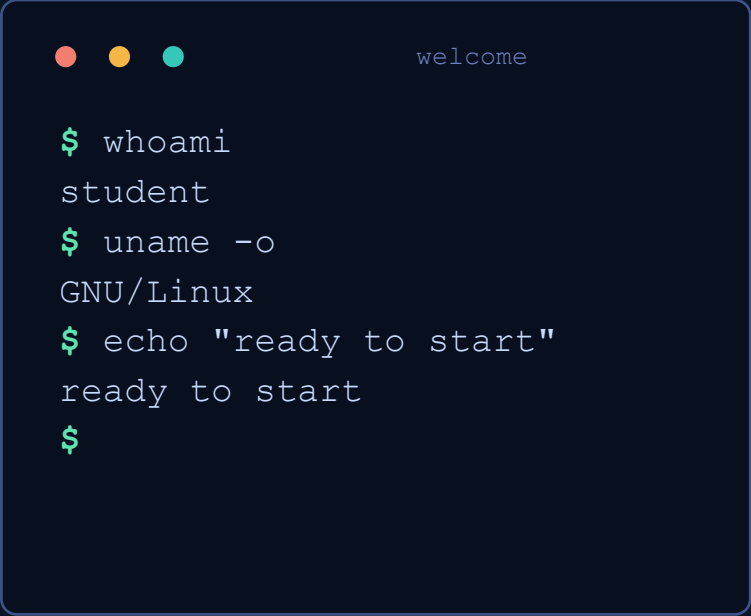


MODULE · THE COMMAND LINE

Meet Linux

A first look at the operating system behind the servers, the cloud, and your VM.

A terminal window mockup with a dark blue background and rounded corners. At the top left are three colored window control buttons (red, yellow, green). At the top right is the text 'welcome'. The terminal shows a series of commands and their outputs: '\$ whoami' followed by 'student', '\$ uname -o' followed by 'GNU/Linux', '\$ echo "ready to start"' followed by 'ready to start', and a final '\$' prompt.

```
$ whoami
student
$ uname -o
GNU/Linux
$ echo "ready to start"
ready to start
$
```

Void · Arch · XUbuntu | *most of our time spent at the terminal*

From Unix to a kernel the world runs on

1969

Unix is born

Ken Thompson, Dennis Ritchie, and others, at Bell Labs build Unix, seeding ideas that still shape systems today.

1983

GNU begins

Richard Stallman starts the GNU project, which sets out to build a complete free operating system and its core tools.

1991

Linus ships a kernel

Linus Torvalds releases the Linux kernel and invites the world to help improve it.

1990s+

Distributions arrive

Kernel plus GNU tools plus choices become Debian, Slackware, Red Hat, and eventually Void, Arch, and Ubuntu.

Strictly speaking, Linux is the kernel; the full system is the kernel plus GNU tools plus a distribution's choices.

WHY IT MATTERS TODAY

You already depend on Linux

100%

of the world's 500 fastest supercomputers run Linux

Billions

of Android phones and devices are built on the Linux kernel



Servers and cloud

Most of the web and cloud infrastructure runs on Linux.



Development

The default home for containers, build pipelines, and modern tooling.



Everywhere else

Routers, cars, TVs, and embedded devices quietly run it too.



Free and open

Anyone can read, run, change, and share the source.

The point is not to memorize these facts; it is to see that the terminal skills you build here transfer nearly everywhere.

Kernel, shell, and you

Hardware

processor, memory, disk

Kernel

talks to the hardware for you

Shell and tools

where your commands live

Let's begin...

typing at the prompt

What is a distribution?

A distribution bundles the Linux kernel with a set of tools, a package manager, and sensible defaults. Void, Arch, and XUbuntu are three such bundles. The core commands you will learn work on all of them.

Some of our CIS students will be building our own Linux distribution!



prompt

```
# anatomy of a prompt
```

```
student@void :~$
```

```
user      host      where you are
```

Next: we open a terminal and start typing.